

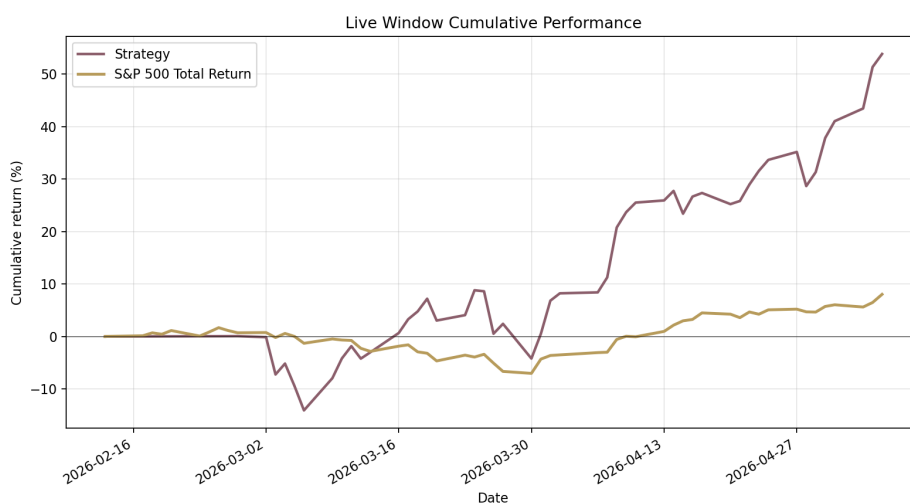
Systematic Long-Short Equity Research

Momentum Selection and Portfolio Accounting

Project - Competition

Investment Research Report

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Abstract

In the Investments & Portfolio Management competition, we sought to outperform the S&P 500 Total Return benchmark with a systematic long-short equity portfolio. The problem was to identify a strategy with attractive net returns while accounting for leverage, trading costs and uncertainty in historical evidence.

We built a research framework linking historical data preparation, strategy comparison and shared portfolio accounting. We selected momentum #1292 through historical comparisons: its 140.37% net return over February 2024-January 2026 was the highest within a five-candidate shortlist. The selected configuration returned 53.85% in the separate competition-period simulation.

Sector attribution reconciles the competition result to account excess return; interaction between sector weights and relative stock returns is the largest separate contribution. Constrained efficient frontiers assess alternative allocations using questionnaire-based risk aversion. Strong simulated returns come with substantial concentration and market exposure. Frontier estimates suggest scope to reduce risk, but their short common sample leaves the optimal weights uncertain.

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1 Introduction

This work was developed for the Investments & Portfolio Management competition, which asked us to manage a systematic long-short portfolio of S&P 500 stocks against the S&P 500 Total Return benchmark. The central problem is to choose a trading rule and assess whether its apparent outperformance compensates for the risks and costs required to obtain it. Historical returns alone cannot answer this question: data conventions, parameter selection, leverage and execution all affect the result.

We address this problem through a shared research framework. Historical prices, constituent membership, sectors and corporate actions feed reproducible strategy searches and a portfolio ledger that records trades, financing and execution costs. Applying the same accounting conventions to momentum, reversal, low volatility and sector momentum allows us to compare their net returns and risks.

Within this framework, we select momentum #1292 from a five-candidate historical shortlist and evaluate it separately over the competition window. Risk measures and sector attribution explain the resulting performance, reconciling excess return to allocation, pure stock selection, interaction and account effects. We also construct constrained efficient frontiers using our questionnaire-based risk preference to study estimated return-risk trade-offs among the selected stocks.

Section 2 sets out the investment objective and risk calibration. Section 3 describes the data and research framework, and Section 4 specifies the momentum rules. Section 5 presents historical selection and comparisons with alternative strategies. Section 6 evaluates competition performance and attribution, while Section 7 examines efficient frontiers. Finally, Section 8 summarizes the findings and priorities for further work.

2 Objective and risk preference

Our objective was to outperform the S&P 500 Total Return benchmark through systematic long-short stock selection. The competition ran from 13 February to 6 May 2026 with \$1 million, S&P 500 stocks, at least ten longs and ten shorts, and a 200% gross exposure ceiling. Our simulations apply its 2% cash and 8% borrowing annual percentage rates, plus a \$2 fee per order and the bid-ask spread.

We calibrate risk preference from ten questionnaire responses, reversing items 2, 3, 8 and 9 as $Q_i^* = 8 - Q_i$:

Item	1	2	3	4	5	6	7	8	9	10
Our response	7	5	1	1	7	5	7	1	3	3
Adjusted score	7	3	7	1	7	5	7	7	5	3

$$\text{BRTI} = \frac{52}{10} = 5.2, \quad q = \frac{5.2 - 1}{6} = 0.7, \quad \lambda = 12.5e^{-0.35(5.2-1)} = 2.8741. \quad (1)$$

This indicates relatively high risk tolerance. The frontier analysis uses $U = \mathbb{E}[R_p] - (\lambda/2)\sigma_p^2$, with annual decimal returns and variance; the calibration does not set strategy leverage.

3 Data and research framework

3.1 Prices and sources

The monthly dataset covers January 2014-January 2026, including benchmark, membership, sector and corporate-action records. We label each month's latest observation by calendar month-end and prioritize Reuters, unadjusted Yahoo `Close`, then a fixed, hash-verified WIKI extract. Missing latest Reuters closes never revive stale quotes. Duplicate Reuters date/security rows merge only if nonmissing

prices and volumes agree; conflicts fail preparation. We neither interpolate prices nor replace historical constituents with today's universe.

Source or evidence	Role and interpretation
Supplied Reuters Yahoo Finance	Primary monthly backtest prices and volumes; reviewed fallbacks fill identified gaps. S&P 500 Total Return benchmark, adjusted daily competition replay prices, and specific backtest or shares requirements.
Historical membership	Effective-dated S&P 500 baskets from the <code>fja05680/sp500</code> history, checked against changes and intervals.
Dated Wikipedia revisions	Historical GICS sector evidence, supplemented by reviewed S&P addition notices only where a new member is absent.
Issuer / SEC evidence	Security events, transaction terms, issuer continuity and reviewed outstanding-share observations.

Table 1: *Data sources and their analytical roles.*

The backtest uses mixed-source *price returns* with reviewed corporate actions but omits ordinary dividends. The competition replay uses adjusted Yahoo prices and reviewed monthly reconstructions, including dividends once in closes or event-return factors. This difference limits comparison with the total-return benchmark. Each result records its price convention and simulation fingerprint.

3.2 Identities and historical coverage

A canonical `Asset_ID` distinguishes ticker changes from conversions into different securities. Mappings are start-inclusive and end-exclusive; overlaps, orphan references and unapproved aliases fail validation. Historical membership baskets, changes and intervals must agree at each decision date. The membership history was reconstructed after the competition.

Sectors use the final Wikipedia revision before 00:00 UTC on each required date. Reviewed S&P notices fill missing new members from their effective dates without overwriting classifications; conflicts fail. Assignments retain GICS codes, source-era labels and references. This reconstruction is not a licensed GICS history.

Structured corporate-action records govern signed entitlements, fractional shares, restricted proceeds and event-date financing. Mandatory delivery incurs no trade fee; a required sale of an off-universe security does. Attribution capitalization uses prices, capitalization factors and reviewed shares, preferring SEC/issuer counts, then event estimates, then consistent Yahoo responses. Missing counts stop preparation; later publications may establish earlier share counts for retrospective attribution.

Manifests and audits record hashes and transformations; raw provider data remain local. Backtest eligibility requires a positive holding-period endpoint price or approved event valuation. Although signals use only lagged history, this availability filter is retrospective and can favor simulated feasibility.

3.3 Portfolio accounting

Strategies receive dated market history and prior targets, then return weights and a decision audit. One ledger handles execution, financing and corporate actions across strategies. Execution can refill the saved ranking without recomputing signals from later observations.

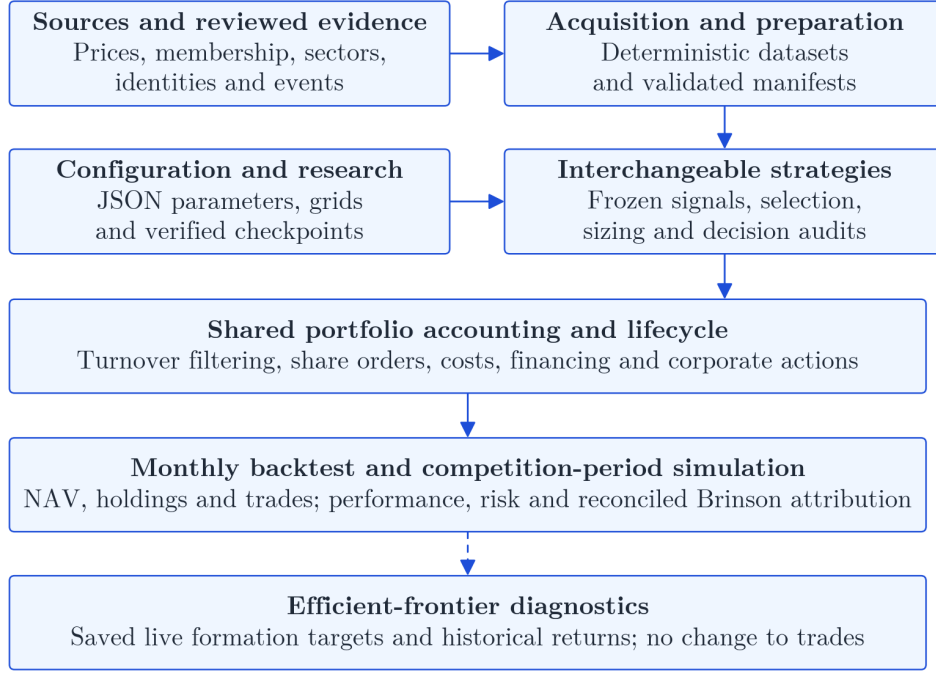


Figure 1: The shared framework. The dashed connection is a diagnostic; frontier allocations do not enter the trading simulation.

For cash C_t , restricted short proceeds R_t , signed units $n_{i,t}$ and prices $P_{i,t}$, the ledger gives NAV, free cash F_t and borrowing L_t :

$$\text{NAV}_t = C_t + \sum_i n_{i,t} P_{i,t}, \quad F_t = C_t - R_t, \quad L_t = \max(-F_t, 0). \quad (2)$$

Positive free cash earns 2%; negative free cash incurs 8%, compounded on Actual/365 days. Execution enforces whole-unit orders, position counts and the gross ceiling. Buys and covers pay the ask; sales and shorts receive the bid, with a modeled half-spread of 1-15 basis points. Each order costs \$2; a sign reversal has two charged legs.

Resolved JSON parameters and assumptions accompany each run. Searches retain rejected, duplicate, history-unready, infeasible and ranked cases separately; candidate numbers are grid-relative. Resume verifies parameters, data, source bytes, runtime versions and output hashes. Tests cover accounting, corporate actions, construction, future-data isolation, identities, checkpoints and attribution.

3.4 Performance measurement

For aligned periodic portfolio, benchmark and risk-free returns r_p, r_b, r_f , we use sample standard deviations ($n - 1$ denominator), with $k = 12$ for monthly history and $k = 252$ for daily competition risk:

$$\sigma_{\text{ann}} = \sqrt{k} s(r_p), \quad \text{Sharpe} = \sqrt{k} \frac{\overline{r_p - r_f}}{s(r_p - r_f)}, \quad (3)$$

$$\text{IR} = \sqrt{k} \frac{\overline{r_p - r_b}}{s(r_p - r_b)}, \quad \text{Treynor} = \frac{k \overline{r_p - r_f}}{\hat{\beta}}. \quad (4)$$

OLS estimates $r_p - r_f = \alpha + \beta(r_b - r_f) + \varepsilon$; reported annual alpha is $k\hat{\alpha}$. Annualized compound return uses NAV growth, whereas these ratios use arithmetic excess means [3, 7]. Maximum drawdown is $\min_t[\text{NAV}_t / \max_{s \leq t} \text{NAV}_s - 1]$.

4 Momentum strategy

Momentum buys recent winners and shorts losers [2]. Our selected configuration, #1292, uses six formation months and a one-month skip. For prepared month-end close $P_{i,t}$, its signal is

$$M_{i,t} = \frac{P_{i,t-1}}{P_{i,t-7}} - 1. \quad (5)$$

We rank eligible stocks in descending order for longs and ascending order for shorts, breaking ties by `Asset_ID`. Eligibility requires finite signals, historical membership, sector records and valid price/event valuations. Initially we take ten stocks on each side, with no overlap. Later decisions retain incumbents within their side’s top fifteen ranks (10×1.5) and fill vacancies in rank order.

4.1 Sizing and trading

With gross exposure $G = 1.5$ and long share $a = 0.7$, target weights are

$$w_{i,t}^{\text{raw}} = \begin{cases} Ga/10 = +0.105, & i \in L_t, \\ -G(1-a)/10 = -0.045, & i \in S_t, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The book is 105% long and 45% short: 150% gross exposure and 60% net long exposure. Positions are equal within each side, with no sector neutrality. We impose no fixed single-position cap.

Targets convert to whole units on each domain’s declared price basis, using the sizing price and NAV. Same-side resizing is suppressed when absolute weight drift is at most 0.05 of NAV. Entries, exits and sign changes still trade; feasibility repairs can override suppression. This five-percentage-point threshold exceeds a short position’s 4.5% target, allowing substantial weight drift.

The historical engine decides, sizes and trades at the labelled month-end close, then values the next interval. The signal is lagged, but sizing and spread inputs use that close. The competition replay instead executes month-end targets at the next scheduled trading-day open.

The rank buffer and resize threshold limit small adjustments, not losses. Concentrated longs, sector imbalance, short squeezes and leverage remain material risks.

5 Historical evaluation

Fixed configurations run continuously over 2015-2023. Ranking uses the 72 monthly returns from 2018-2023; six annual slices assess variation without resetting capital or reselecting parameters. Longer estimators can delay activation beyond the nominal 2014 warm-up.

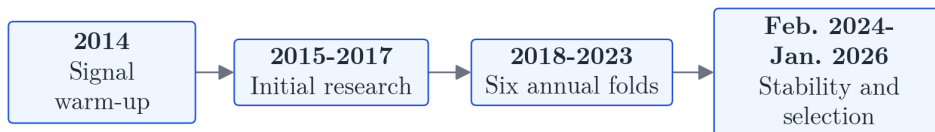


Figure 2: *Historical windows. The test portfolio starts on January 31, 2024 and produces 24 monthly returns.*

We ranked 5,760 momentum configurations by net return, varying formation, position counts, sizing, exposure, buffers, turnover thresholds and sector neutrality. Candidate #1292 ranked fourth, then led

the top-five shortlist in the test window ([Table 2](#)).

Candidate	Validation return rank	Validation return	Test return	Test Sharpe	Test drawdown
#1436	1	+78.02%	+108.11%	1.25	−16.87%
#1712	2	+74.47%	+95.34%	1.17	−16.23%
#1724	3	+74.44%	+103.37%	1.24	−15.40%
#1292	4	+73.79%	+140.37%	1.28	−16.07%
#1424	5	+70.72%	+101.52%	1.20	−16.69%

Table 2: *The return-focused momentum shortlist. Validation covers January 2018–December 2023; the test window covers February 2024–January 2026. Returns are cumulative and net of modeled costs.*

In the test window, #1292’s volatility was 38.27%, versus 10.63% for the benchmark. January 2026 supplied 48.37% of cumulative profit; the preceding 23 months still returned 72.48%, versus 44.94%. Annualized compound return was 55.04%, against 21.26%; beta was 1.49, Treynor 0.329 and Information Ratio 0.881. Annualized CAPM alpha was 22.17% ($p = 0.439$), not statistically significant at the 5% level.

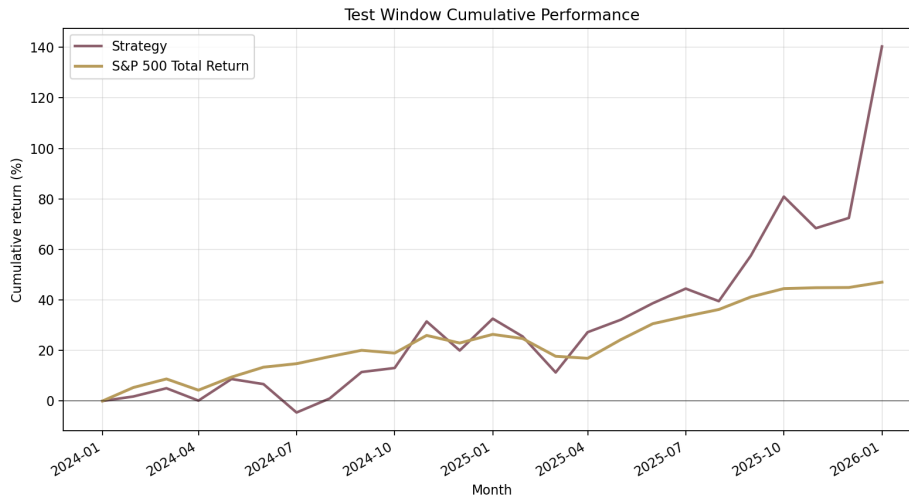


Figure 3: *Momentum #1292: cumulative net performance, Feb. 2024–Jan. 2026.*

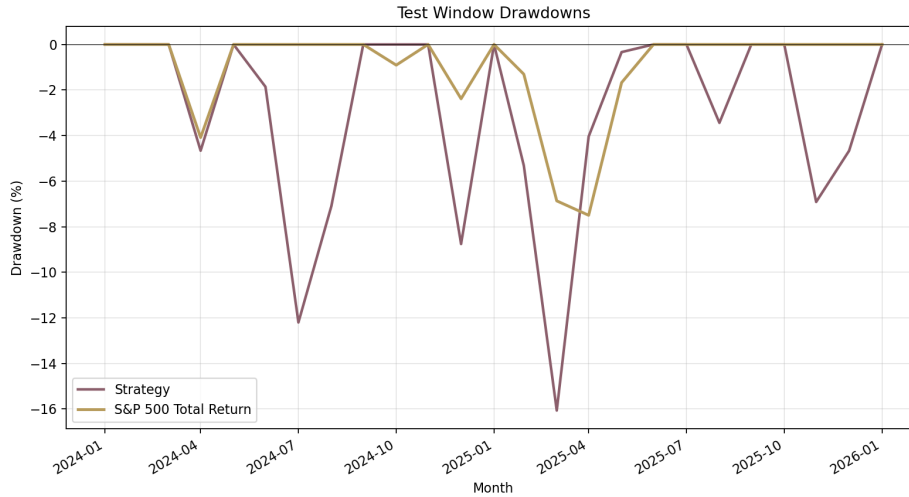


Figure 4: *Momentum #1292: month-end drawdown, Feb. 2024–Jan. 2026.*

5.1 Costs and alternatives

The test run placed 371 orders, costing \$742 in fees, \$6,837 in spreads and \$17,693 in loan interest. Return before fees and spreads, but after financing, was 142.57%, versus 140.37% net. Monthly turnover averaged 1.165 times pre-trade equity, counting absolute purchases and sales at reference prices. Low/base/high spread cases returned 140.68%/140.37%/140.06%; these vary only the modeled spread coefficient.

We compared three other families on the same validation dates and accounting basis (Table 3); drawdown uses the initial wealth level and monthly marks.

Strategy	Ranked candidates	Candidate	Net return	Sharpe	Drawdown
Momentum	5,760	#1292	+73.79%	0.420	−26.36%
Reversal	1,152	#1123	+50.79%	0.327	−31.50%
Low volatility	1,152	#598	+15.49%	0.094	−22.75%
Sector momentum	967	#881	+69.34%	0.556	−16.42%

Table 3: *Selected validation results for January 2018–December 2023. The alternative rows show each family’s cumulative-return leader; momentum shows our selected configuration. Sector momentum attempted 1,296 compatible configurations, of which 329 were execution-infeasible.*

Reversal buys prior-month losers and shorts winners. Return and Sharpe leader #1123 turned over 132.85 times equity and paid \$66,879 in fees and spreads; median grid return was −6.30%.

Low volatility buys the least volatile stocks and shorts the most volatile, using 60 months with at least 24 observations. Only 39 of 1,152 configurations profited; median return was −36.94%. Sharpe leader #599 scored 0.095 and lost money in every year from 2019 through 2023.

Sector momentum ranks equal-weight sector returns and trades whole baskets; the leaders use three long and three short sectors. Sharpe leader #763 returned 45.81%, with Sharpe 0.598 and drawdown 7.23%. Our return-first comparison favored momentum. Infeasible configurations failed position counts under scaled whole-share execution and receive no investment returns.

Monthly trend ranks price relative to its monthly moving average. Its strategy and grid are implemented, but the grid search was not run; we draw no empirical performance conclusion.

Shared samples and related parameters create selection bias; annual slices are not independent economic regimes. Their returns appear in Table 4.

Year	Momentum #1292	Reversal #1123	Low volatility #598	Sector momentum #881
2018	−18.89%	+1.69%	+21.15%	−9.60%
2019	+3.09%	+28.91%	−1.65%	+9.13%
2020	+30.07%	+14.73%	−0.53%	+22.94%
2021	+14.49%	+12.95%	−1.31%	+14.29%
2022	+1.62%	−12.76%	+0.33%	+18.01%
2023	+37.35%	+1.75%	−1.59%	+3.52%

Table 4: *Annual net returns within the continuous 2018–2023 validation run.*

6 Competition simulation and attribution

We simulate #1292 over the competition dates. The account stays in cash until month-end. Formation/execution dates are February 27/March 2, March 31/April 1 and April 30/May 1, with ten longs

and ten shorts in each invested interval. Sizing uses the preceding close; overnight moves, integer shares and retained positions explain deviations from the formation weights.

The account ends at \$1,538,521: a 53.85% net return against 8.04% for the benchmark, or 45.82 percentage points of excess return. The gain follows a March decline and sharp recovery in April and early May.

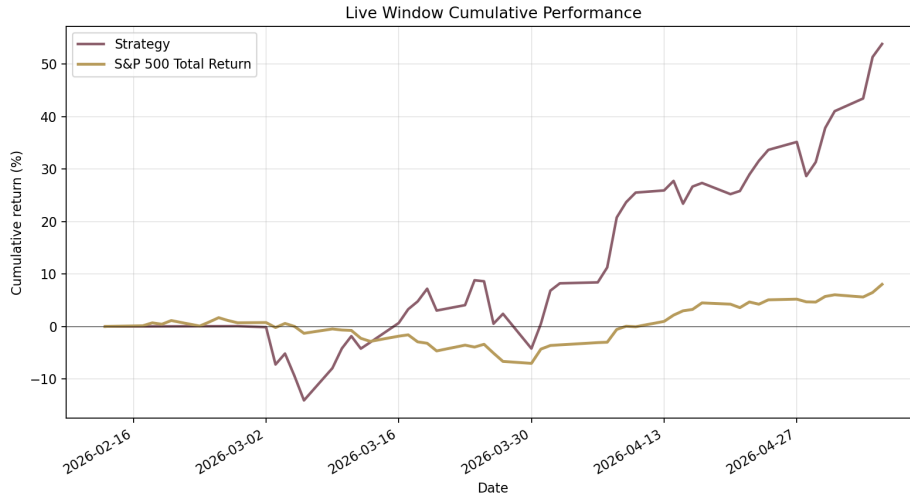


Figure 5: *Net momentum simulation and S&P 500 Total Return benchmark, February 13 open-May 6 close, 2026, including the initial cash phase.*

6.1 Risk and costs

Risk statistics use 56 close-to-close returns, including the cash phase, at 252 observations per year. The initial open-to-close move enters cumulative performance only. Risk-free accrual uses the 2% Actual/365 rate; metric formulas are in [Section 3.4](#).

Metric	Momentum	S&P 500 TR
Annualized volatility (%)	52.28	15.11
Sharpe ratio	3.94	2.24
Market beta	1.99	1.00
Treynor ratio	1.04	0.34
Information Ratio	3.80	-
Tracking error (%)	45.33	-
Annualized CAPM alpha (%)	138.98	0.00

Table 5: *Competition-period risk metrics. Rates are annualized.*

Beta near two indicates substantial market exposure. Annualizing the 53.85% gain over 82 calendar days gives 580.52%. Annualized CAPM alpha was 138.98% ($p = 0.139$), not statistically significant at the 5% level. The short sample also limits the reliability of risk ratios.

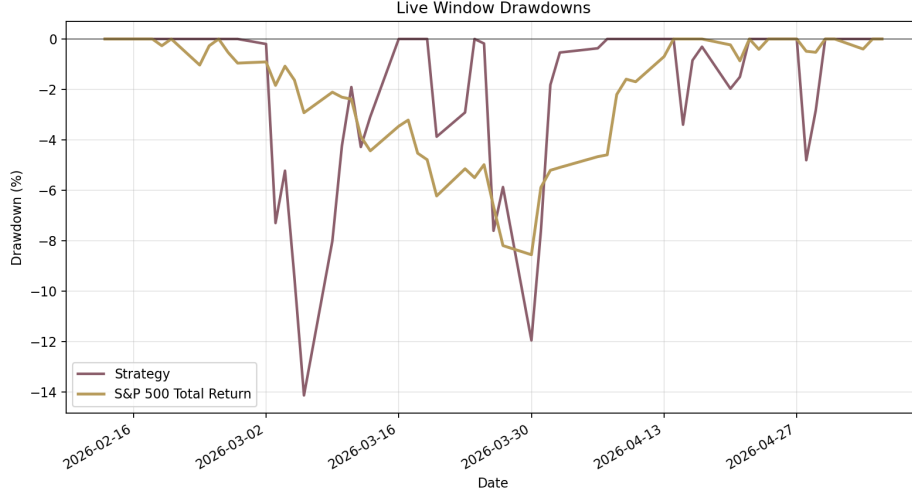


Figure 6: Daily drawdowns, including the initial cash phase. Maximum drawdown is -14.14% for momentum and -8.56% for the benchmark.

The 36 orders cost \$72 in fees and \$498.21 in spreads. Cash earned \$931.92 and borrowing cost \$430.78. Low/base/high spread cases returned $53.86\%/53.85\%/53.84\%$.

The full trade log retains decision and execution dates, identities, targets, applied rules, rounding, suppression, constraint overrides and cash effects. Table 6 gives four entries; signals are six-month decimal returns and prices are adjusted-unit execution references before spreads.

Date	Security	Rule	Units traded	Signal	Price (\$)	Spread (\$)
2 Mar	ALB	Enter long	593	1.5361	172.0708	21.12
2 Mar	COIN	Enter short	-256	-0.4845	172.5000	6.64
1 Apr	INTC	Exit long	-2,304	0.8731	45.0000	13.95
1 May	LRCX	Exit long	-451	0.6002	254.6898	16.76

Table 6: Selected competition-simulation trade-log entries.

6.2 Attribution and reconciliation

We apply Brinson-Fachler attribution to each sleeve, normalizing absolute weights and reversing both stock and benchmark returns for shorts [1, pp. 12-14]. The reconstructed benchmark total is $\bar{r}_t^b = \sum_s w_{s,t}^b r_{s,t}^b$, using the same sleeve-specific signs. For sector s in interval t , allocation, pure selection and interaction are

$$\begin{aligned} A_{s,t} &= (w_{s,t}^p - w_{s,t}^b)(r_{s,t}^b - \bar{r}_t^b), \\ S_{s,t} &= w_{s,t}^b(r_{s,t}^p - r_{s,t}^b), \quad I_{s,t} = (w_{s,t}^p - w_{s,t}^b)(r_{s,t}^p - r_{s,t}^b). \end{aligned} \tag{7}$$

For an unheld sector, we set selection and interaction to zero without assigning a realized portfolio return. Sector effects differ from raw active contributions by $(w_{s,t}^p - w_{s,t}^b)\bar{r}_t^b$; these adjustments cancel across normalized sectors. Multiplying each sleeve's effects once by its positive gross exposure expresses them relative to post-trade NAV.

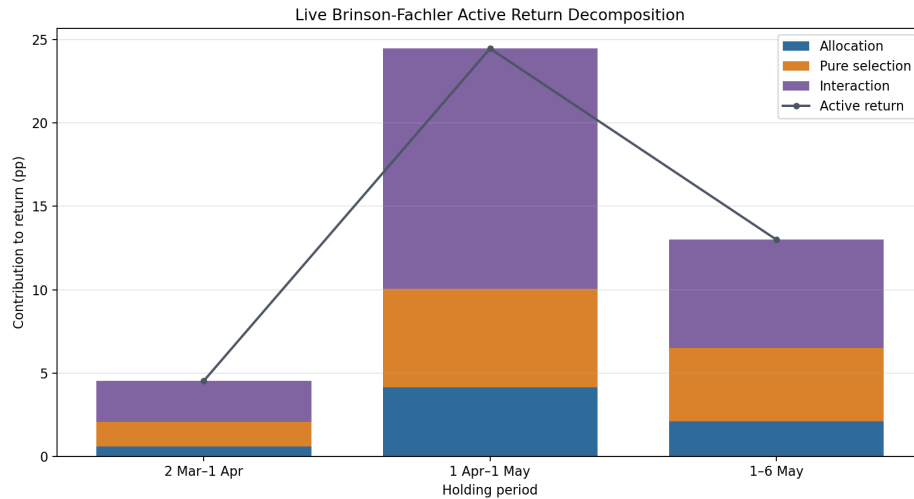


Figure 7: *Brinson-Fachler allocation, pure selection and interaction over the three invested intervals, before account adjustments and compounding.*

The reconstructed benchmark covers all 503 eligible constituents in each interval, with complete prices, shares and sectors and weights summing to one. It returns 9.13%, versus 8.04% for the official index; the largest interval gap is 1.06 percentage points (Figure 8). Attribution retains this discrepancy explicitly.

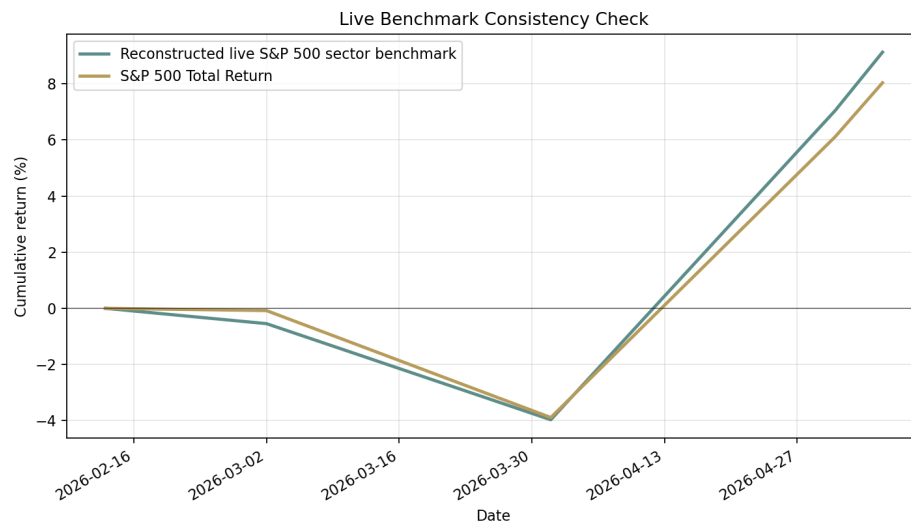


Figure 8: *Reconstructed sector benchmark versus the official total-return index, February 13 open-May 6 close, 2026. Their difference enters account attribution.*

To explain the account's net excess return, we first express all contributions as percentages of its value at the start of each interval. We add two adjustments to allocation, pure selection and interaction: one for our net stock exposure differing from a 100%-invested benchmark, and one for the gap between reconstructed and official benchmark returns. Cash interest increases return; borrowing, fees and spreads reduce it.

These contributions $e_{k,t}$ sum to each interval's excess $e_t = R_t^p - R_t^b$. Because returns compound, summing interval effects alone does not explain cumulative excess. We multiply each by a linking factor L_t : portfolio growth before interval t times benchmark growth after it. The resulting totals E_k satisfy

$$\begin{aligned} L_t &= \prod_{j<t} (1 + R_j^p) \prod_{j>t} (1 + R_j^b), & E_k &= \sum_t L_t e_{k,t}, \\ \sum_k E_k &= \sum_t L_t e_t = R_{\text{cum}}^p - R_{\text{cum}}^b. \end{aligned} \tag{8}$$

Including the initial cash interval, the summed effects match each interval's measured excess return to numerical precision.

Linked contribution	Percentage points
Sector allocation	7.8277
Pure stock selection	13.7421
Interaction	26.5692
Net-exposure adjustment	−3.5176
Benchmark-reconstruction adjustment	1.1995
Cash interest	0.1008
Borrowing interest	−0.0455
Fixed fees	−0.0077
Spread costs	−0.0532
Total net account excess	45.8152

Table 7: Contributions to compounded net account excess, in percentage points. Displayed effects are rounded independently.

The chart's 42.00 percentage points simply add stock effects across invested intervals. On a common NAV basis with compounding, the three stock effects contribute 48.14 points. Account adjustments bring this to 45.82 points. Interaction is the largest separate effect: 24.33 points come from the long sleeve and 2.24 from the short sleeve. Long Information Technology alone contributes 21.54 interaction points, reflecting its roughly 80% sleeve weight versus 29-31% in the benchmark and stronger selected-stock returns, especially in April. Pure selection contributes 13.74 points. This decomposition describes realized concentration and relative returns; it does not establish causality or persistent stock-picking skill.

7 Efficient frontier

We compare the momentum allocation with mean-variance optima using all twenty nonzero targets at first formation, February 27, 2026. Reference weights use the original sizing NAV of \$1,000,767.40.

7.1 Estimation and constraints

We retain only months with returns for all twenty stocks. SNDK began trading in February 2025 after separating from Western Digital [8], so our sixty-month lookback yields only twelve common monthly returns, March 2025-February 2026. Requests for 12, 24 or 36 months yield the same sample. February's month-end label represents the February 27 close. We compute adjacent-month returns before removing incomplete intervals, without filling prices, bridging gaps or stitching different securities.

Following Markowitz [5], annual arithmetic means and covariance are

$$\hat{\mu} = 12\bar{r}, \quad \hat{\Sigma} = 12 \left[(1 - \delta)S_{ML} + \delta \frac{\text{tr}(S_{ML})}{20} I \right]. \quad (9)$$

Ledoit-Wolf shrinkage uses centered covariance normalized by n , a spherical target and fitted $\delta = 0.3095$ [4, 6]. It raises covariance rank from eleven to twenty, with condition number 26.03.

Both feasible sets preserve stock directions and sleeve totals:

$$\sum_{i \in \mathcal{L}} w_i = 1.05, \quad \sum_{i \in \mathcal{S}} w_i = -0.45. \quad (10)$$

Bounded weights stay between 0.5 and 2 times their original magnitudes, retaining ten longs and ten shorts. *Relaxed* weights can be zero or reach their sleeve total, allowing portfolios below the competition's position minimum. Neither set imposes sector neutrality.

We trace each efficient branch from its global minimum-variance portfolio (GMV) using one hundred return targets, and separately maximize

$$U(\mathbf{w}) = \mathbf{w}^T \hat{\mu} - \frac{\lambda}{2} \mathbf{w}^T \hat{\Sigma} \mathbf{w}, \quad \lambda = 2.8741. \quad (11)$$

Indifference curves satisfy $\mu_p = U_0 + \lambda \sigma_p^2 / 2$ in annual decimal units. An independent linear certificate checks feasibility and the first-order optimality gap.

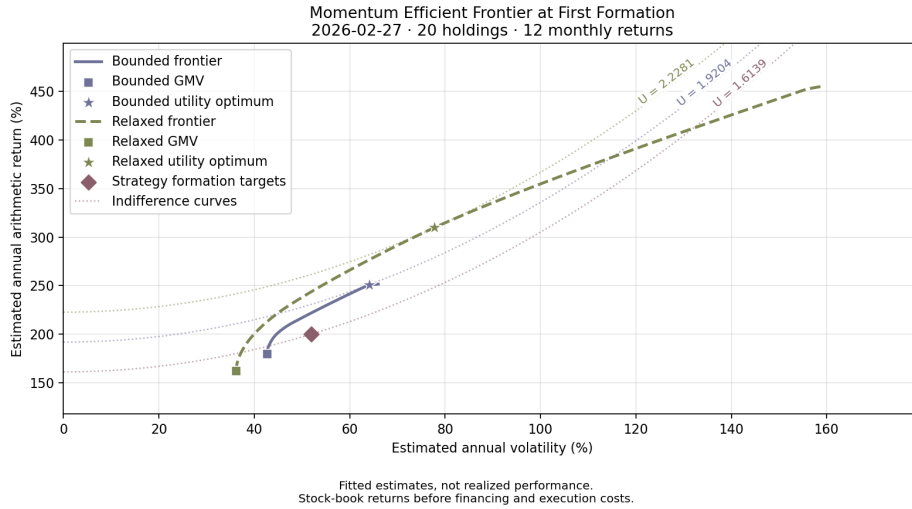


Figure 9: First-formation frontier and indifference curves for all twenty targets.

7.2 Allocation and sensitivity

At the formation portfolio's estimated return, bounded reallocation lowers volatility from 51.99% to 44.66%; the relaxed minimum is 39.95%. Both bounded optima retain ten longs and ten shorts, with a 21% largest position. The relaxed GMV holds eight longs and four shorts, with a 25.41% largest position; GMV utilities are 1.538 bounded and 1.432 relaxed. The relaxed utility optimum concentrates into three longs and two shorts, with 43.02% in the largest position.

The large fitted returns reflect short-sample annualization and leverage. The sample also overlaps parameter selection, so these are in-sample allocation comparisons. Table 8 tests estimation and

constraint choices.

Estimation or constraint case	Return (%)	Vol. (%)	Utility
Primary Ledoit-Wolf	251.16	64.14	1.920
Means shrunk 50%	121.89	47.70	0.892
20% diagonal covariance shrinkage	238.35	65.78	1.762
50% diagonal covariance shrinkage	248.25	62.05	1.929
Tighter bounds: $0.75\text{-}1.5\times$	225.68	57.81	1.776
Wider bounds: $0.25\text{-}3\times$	285.63	73.49	2.080
Leave one month out: range	206.53-283.17	51.09-68.55	1.682-2.292

Table 8: *Sensitivity of the bounded utility optimum. Rows use the same primary sample except the explicitly labelled leave-one-month-out perturbations.*

Mean shrinkage pulls each stock halfway toward the cross-sectional mean. The diagonal cases use unbiased $(n - 1)$ sample covariance, unlike the primary estimator’s n normalization and spherical target; both features therefore change. Leave-one-month-out ranges are perturbations, not confidence intervals. Shrinkage improves conditioning, but expected returns remain poorly estimated.

8 Conclusions

The competition posed a practical investment problem: pursue benchmark outperformance through a systematic long-short portfolio while making its risks and costs explicit. We addressed it with a framework connecting historical data preparation, comparable strategy searches and shared portfolio accounting. Trade audits and reconciled attribution connect the reported returns to the underlying decisions and account movements.

Our return-focused comparison selected momentum #1292 over more defensive alternatives, available both within momentum and through sector momentum. We evaluated the selected configuration on the competition window, where it outperformed the benchmark after modeled costs. Interaction between sector weights and relative stock returns was the largest attribution effect, led by the concentrated long Information Technology sleeve. The short window and statistically insignificant CAPM alpha in both the test and competition windows limit the evidence for persistent risk-adjusted outperformance. The frontier identifies reallocations with lower estimated volatility at the same estimated return, based on twelve monthly observations. We have yet to evaluate a trading strategy using those optimized weights.

Our next step would be to extend evaluation of the fixed strategy beyond the competition window, comparing it with prespecified defensive alternatives under wider spread and borrowing-cost assumptions. We would also test whether rolling frontier allocations improve net performance. To reduce reliance on the short sample, we would pull extreme estimated returns toward a common average and investigate methods that retain older observations for stocks with longer histories.

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